An equilateral triangle in space

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Let ABC be an equilateral triangle with side a. Let $\mathcal{C}_A$ be the circle lying in the perpendicular bisector plane of $[BC]$, centred at the midpoint of $[BC]$, with a fixed radius r. Define $\mathcal{C}_B$ and $\mathcal{C}_C$ likewise on the sides $[AC]$ and $[AB]$. Let A' be a point of $\mathcal{C}_A$ such that there are two points B' and C'' of $\mathcal{C}_B$ at distance a from A'. Let C' be the point of $\mathcal{C}_C$ symmetric to C'' with respect to the perpendicular bisector plane of $[BC]$. Show that $A'B'C'$ is an equilateral triangle.